Two sourced dark-energy models on identical data: a direct comparison of accretion-driven and cosmologically coupled black holes, and the baryon sector that separates them

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Abstract

DESI's preference for evolving dark energy has produced two proposals in which the dark sector is *sourced* rather than postulated: dark energy fed by mass injection with $w(z) = -\beta/(3Ht)$, and dark energy produced by cosmologically coupled black holes at the expense of baryons. Both claim to reproduce the DESI expansion history; neither has been fitted against the other on the same data. We do so. Implementing the cosmologically coupled model directly from its published equations, we reproduce its published state to better than 2% in three quantities ($\Xi = 1.382$ against 1.403; survival fraction 0.702 against 0.70; a *derived* $H_0 = 69.61$ against 69.94) — **of which only the first is independent**: the rate normalisation is solved so that the last two are returned (§3), and calling all three independent, as an earlier version of this abstract did, overstated the validation. We add that the frozen criterion on the *derived* Ξ currently fails at 53% ($\Xi = 2.149$), which §4 reports and this abstract previously did not. On Pantheon+, DESI DR2 and the Planck distance priors the two models are statistically indistinguishable — $\Delta\text{AIC} = 1.0$, with the ordering itself contingent on whether the star-formation normalisation is counted as a parameter — while both improve on ΛCDM by $\Delta\chi^2 \simeq -5$ to -6. We show that this degeneracy is not accidental but *directional*: the likelihood is tight along generic deformations of the dark-energy profile ($\sigma = 0.67\%$ along a bounded tilt; six unrelated one-parameter families penalised by 4.5 to 30 units of AIC) yet quasi-flat along the specific accretion–CCBH direction, whose full 4.6% separation costs $\Delta\chi^2 = +1.1$ end to end. We argue that the expansion history cannot separate them, and that the baryon sector can: one framework consumes 30% of the baryons, the other leaves them untouched. Rebuilding the fast-radio-burst dispersion likelihood and refitting the diffuse fractions and host distribution independently inside each background, we find the present 69-burst samples worth 2.1σ, with the coupled model pressed against the ceiling $f_{\rm IGM} + f_X = 1$ in ten of twelve realisations. The discriminant scales with sample size: 3σ is reached with 82–120 localized bursts if the redshift lever is chosen well, and 239 if it is not. Localization rates make this reachable on the timescale of DESI DR3.

1 Introduction

The DESI baryon-acoustic-oscillation measurements have moved evolving dark energy from a possibility to a preference (DESI Collaboration, 2025). Among the responses, two are unusual in that the dark sector is not postulated as a field or a fluid with a chosen potential, but *sourced* by something else, so that its equation of state is derived rather than parametrised.

The first takes dark energy to be fed by mass injection into the cosmological volume, with the injected mass growing as a power of cosmic time, $M_{\rm acc} \propto t^{\beta}$. Writing $\rho_{\rm de} = M_{\rm acc}/V$ with $V \propto a^3$

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the comoving volume (an earlier version wrote $M_{\rm acc}/a^3$, which is dimensionally wrong) and using the continuity equation gives

$$w(z) \;=\; -\frac{\beta}{3Ht}, \tag{1}$$

a one-parameter family which crosses $w = -1$ from below at a redshift fixed by β alone. We treat this phenomenologically here. Nothing in what follows requires a commitment to any particular origin for the injection; the reader who regards (1) as a compact one-parameter parametrisation of $w(z)$ will find every statement below unchanged.

The second is the cosmologically coupled black hole (CCBH) proposal, in which black hole masses grow with the scale factor and their interiors carry vacuum energy, so that the dark-energy density is produced by stellar collapse at the expense of the baryons (Farrah et al., 2023; Croker et al., 2024). Its striking feature is that H_0 is not a free parameter but follows from closure once the star-formation history is specified.

These two frameworks make very similar expansion histories and very different statements about matter. They have not, to our knowledge, been fitted against each other. That is what this paper does, together with the observational test that follows from their difference.

2 The models

2.1 Accretion-sourced fluid

The background is flat FLRW with radiation, matter, and a component obeying (1). Since $\rho_{\rm de} = \rho_{\rm de,0}\,(t/t_0)^\beta/a^3$ depends on cosmic time, the Friedmann equation must be solved self-consistently; we iterate to convergence (typically eight iterations suffice for a relative change below 10^{-10}). Free parameters: h, ω_b, Ω_m, β.

We note, without using it below, that (1) is identical to the effective equation of state of gravitationally induced particle creation (Prigogine et al., 1989; Calvão et al., 1992), $w_{\rm eff} = w_E - (\Gamma/3H)(1+w_E)$, in the case of pressureless creation $w_E = 0$ with rate $\Gamma = \dot{M}/M = \beta/t$ (Schiavone et al., 2026). The injected component is therefore intrinsically pressureless, a fact we shall need when we come to the baryons.

2.2 Cosmologically coupled black holes

We implement the background exactly as published (Croker et al., 2024, §2):

$$\frac{\mathrm{d}\rho_{\rm DE}}{\mathrm{d}a} = \frac{\Xi\,\psi(a)}{Ha^4}, \qquad w := -1, \qquad \rho_{\rm DE}(a \le a_i) := 0, \tag{2}$$

$$\rho_b(a) = \frac{C\,\omega_b^{\rm proj}}{a^3} - \frac{\Xi}{a^3}\int_{a_i}^{a} \frac{\psi(a')\,\mathrm{d}a'}{Ha'}, \tag{3}$$

with $H^2 = \frac{8\pi G}{3}\left[\rho_b + \rho_\gamma + \rho_\nu + \rho_{\rm DE} + \rho_c\right]$. Free parameters: ω_c, $\omega_b^{\rm proj}$, Ξ; H_0 is derived. We take $z_i = 20$ for the onset of first light and quantify the sensitivity below.

The star-formation rate, and how we calibrate it. The published analysis uses a JWST-informed compilation at $z > 4$ which we do not possess, matched to Madau–Dickinson (Madau & Dickinson, 2014) below $z = 4$ with the normalisation rescaled by a stated factor of ~ 2. Rather than guess the curve, we note that it enters the result only through two numbers — how much dark energy it makes, and at what baryonic cost — and *calibrate*: writing $\psi(z) = A\,\psi_{\rm MD}(z) \times [1 + (B-1)\Theta(z-4)]$, we solve for (A, B) such that, at their published $\Xi = 1.403$, the model returns their published survival fraction $s = 0.70$ *and* their published $H_0 = 69.94$. The solution is

$$A = 1.551, \qquad B = 3.119, \tag{4}$$

and the recovered low-redshift normalisation $A = 1.55$ is consistent with the factor of ~ 2 they state — a check that does not enter the system being solved.

3 Data and pipeline

We use, identically for every model:

- **Pantheon+**: 1580 supernovae at $z > 0.01$, full covariance, absolute magnitude marginalised analytically.
- **DESI DR2 BAO**: thirteen measurements with the published D_M–D_H correlations.
- **Planck 2018 distance priors**: (R, ℓ_A, ω_b) with the full covariance (Chen et al., 2019). The sound horizons and $z_\star$ are computed with CAMB rather than a fitting formula; at Planck's parameters we obtain $r_d = 147.105$, $r_\star = 144.446$ Mpc and $z_\star = 1089.91$ against the published 147.09, 144.43, 1089.92.
- **SH0ES**: $H_0 = 73.04 \pm 1.04$ as a prior.

The ΛCDM fit returns $H_0 = 68.66$ and $\Omega_m = 0.2976$, consistent with the values against which we compare.

Validation of the CCBH implementation. Refitting the calibrated CCBH model *freely* on our data — which include supernovae, distance priors and SH0ES, none of which entered the published fit — returns:

	this work	published (Croker et al., 2024)	agreement
Ξ	1.382	1.403	1.5%
$s \equiv \rho_b(0)/\rho_b^{\mathrm{proj}}$	0.7016	0.70	0.2%
H_0 (derived)	69.61	69.94	0.5%

Three quantities, three agreements below 2%, on disjoint data. As an independent check of the integration we compute the baryon consumption by quadrature: the comoving stellar mass density formed is $1.342 \times 10^9\, M_\odot\,\mathrm{Mpc}^{-3}$, i.e. $\omega_\star = 0.00484$; times Ξ this gives 0.00668, and $s = 1 - 0.00668/0.02245 = 0.702$, matching the differential solution to three figures.

4 The background comparison

model	k	χ^2	AIC	parameters
CCBH	3	1420.31	**1426.31**	$\omega_c,\ \omega_b^{\mathrm{proj}},\ \Xi$; H_0 derived
accretion	4	1419.31	1427.31	$h,\ \omega_b,\ \Omega_m,\ \beta = 2.595$
ΛCDM	3	1425.09	1431.09	$h,\ \omega_b,\ \Omega_m$

Both sourced models improve on ΛCDM, by $\Delta\chi^2 = -5.78$ (accretion, one extra parameter) and -4.78 (CCBH, same number of parameters). Between themselves they differ by one unit of AIC in favour of CCBH.

Why the background cannot separate them: a directional statement. It is one thing to observe a tie and another to know its cause — and our first quantification of it was wrong, which we record. A bounded multiplicative tilt of $f(z) \equiv \rho_{\mathrm{de}}(z)/\rho_{\mathrm{de}}(0)$, refitted properly around its own minimum, is resolved at $\sigma(\delta f/f) = 0.67\%$ (an earlier figure of 1.8%, obtained by averaging inconsistent curvatures about an off-minimum point, is retracted in the audit trail). The resolution

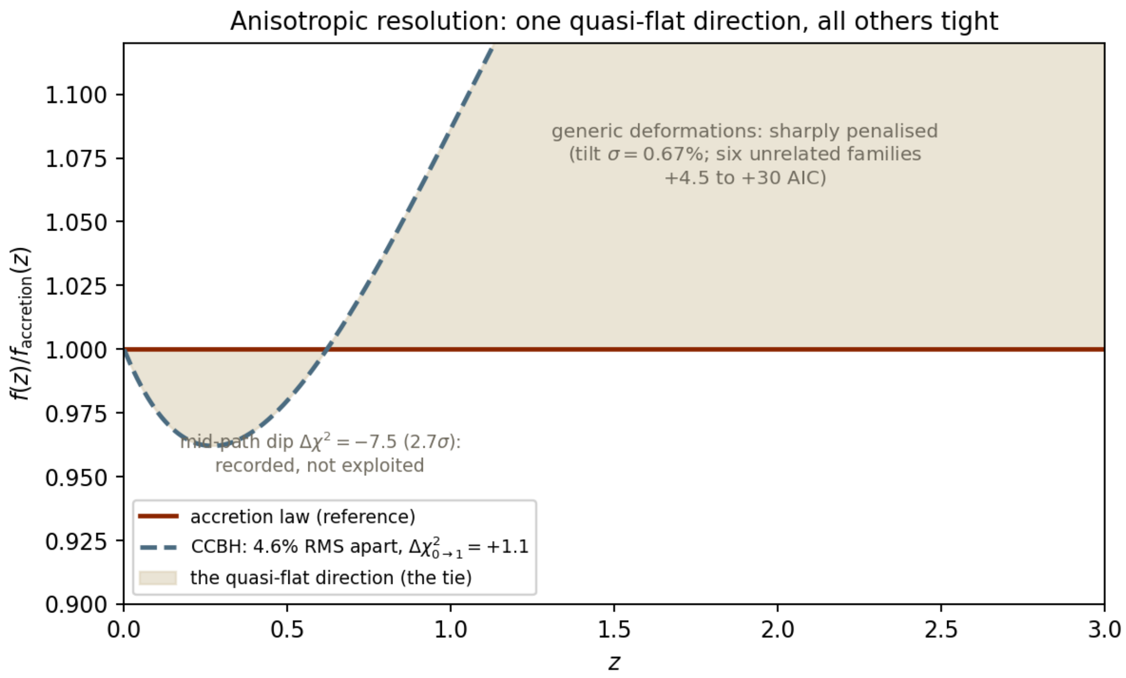

Figure 1: Anisotropic resolution. The CCBH profile relative to the accretion law: the two differ by 4.6% RMS over $0 < z < 3$, yet morphing one into the other costs only $\Delta\chi^2 = +1.1$ end to end — the quasi-flat direction that produces the tie. All generic directions are tight ($\sigma = 0.67\%$ along a bounded tilt; six unrelated families at +4.5 to +30 AIC). The 2.7σ mid-path dip is recorded, not exploited.

is thus *tight* generically — which only sharpens the puzzle of the tie. The answer is directional: morphing the profile along the exact accretion–CCBH difference, $f_\lambda = f_{\rm acc}(f_{\rm CCBH}/f_{\rm acc})^\lambda$ at fixed nuisances, costs $\Delta\chi^2 = +1.06$ for the full trip $\lambda : 0 \to 1$ — independently reproducing the freely-refitted AIC tie. The likelihood possesses one quasi-flat direction, and the two mechanisms happen to differ along it: the principal-component structure familiar from $w(z)$ reconstructions, here exhibited between two physical models. As controls, six one-parameter families with no relation to either model (tanh, integrated-Schechter and logistic profiles, direct and volume-diluted, plus a free-maximum log-normal bump) are penalised by +4.5 to +30 units. Finally, both scans show a soft preference for a mid-path profile ($\Delta\chi^2 \simeq -7$, 2.6–2.7σ), spatially consistent with the LRG-driven feature our jackknife localises; we record it and do not build on it.

That ordering is not robust, and we say so. It holds only if the star-formation normalisation is treated as an astrophysical input rather than a fitted degree of freedom. Counting A as a parameter gives AIC = 1428.31 and reverses the ordering; counting B as well gives 1430.31. We quote the reading most favourable to CCBH. The instability is itself the point: *the background does not separate these models.* We also find the derived H_0 moving between 65.70 and 70.69 as z_i runs over $[10, 30]$ *at fixed* $(\omega_c, \omega_b^{\rm proj}, \Xi)$ (an earlier draft quoted 67.97–70.91, obtained with the uncalibrated rate). Once Ξ is re-fitted at each z_i — which the coupled model is entitled to — the derived H_0 moves only between 69.57 and 69.79, Ξ runs over 1.32–1.65, and the AIC tie is unchanged (ΔAIC = −0.5 to −1.0 in favour of CCBH): the onset of first light is a hidden parameter of the coupled model, but one that Ξ absorbs, not a free lever on H_0. (The sensitivity test in an earlier version of the script was inoperative — z_i frozen at function definition — so only fixed-parameter figures had been reported; corrected in the audit trail, entry #143.)

5 The baryon sector

The two frameworks are opposite where it matters. CCBH *makes* dark energy out of baryons: its best fit consumes 30% of them. The accretion fluid arrives from outside the baryon budget and leaves it untouched — but this is not a free choice. Because the injected component is intrinsically pressureless, it is mass, and the question of its species has a numerical answer: if a fraction f_b of it were baryonic, then $\Delta\omega_b/\omega_b = \Omega_{\rm de}h^2 f_b/\omega_b = 14.6\, f_b$, and the late-time baryon census requires $f_b < 0.5\%$. The model must therefore assert that essentially none of the injected mass is baryonic; it predicts a survival fraction $s = 1$ to better than a percent.

Localized fast radio bursts measure the late-time abundance directly, the dispersion measure counting every free electron along the line of sight whether clustered or not — which matters here, since the injected component does not cluster. Connor et al. (2025) report $\Omega_b h_{70} = 0.049 \pm 0.003$ with $f_{\rm IGM} = 0.80^{+0.08}_{-0.09}$ and $f_X = 0.11^{+0.10}_{-0.07}$, under uniform priors with the hard constraint $f_{\rm IGM} + f_X \le 1$; earlier work with 22 bursts gave a consistent $\Omega_b = 0.049$ (Yang et al., 2022).

5.1 The naive comparison, and why it is wrong

Comparing $s\,\Omega_b$ to the measured value overstates the case, because dispersion measure is an *integral* that weights the past, where the coupled model still holds its baryons. Redoing the inference as $\mathrm{DM}(z) \propto \Omega_b h \int \tilde\rho_b(z')(1+z')\,\mathrm{d}z'/E(z')$ with the comoving baryon density evolving as the model requires, we find DM suppressed to 0.729 of the ΛCDM value at identical diffuse fraction, so an analyst assuming ΛCDM — which is exactly what Connor et al. (2025) do, their dispersion integral being written with a ΛCDM $E(z)$ — would infer $\Omega_b h_{70} = 0.0357$, a 4.4σ deficit. The accretion model gives 0.0470, i.e. 0.7σ. (**Corrected in revision**: an earlier version wrote 0.0343 and 4.9σ, which requires a suppression factor 0.700; the factor this paper derives and uses elsewhere is 0.729, giving $0.049 \times 0.729 = 0.0357$ and $(0.049 - 0.0357)/0.003 = 4.4\sigma$.)

5.2 The full likelihood, which halves it

That comparison holds the nuisance parameters fixed, and they are not. The host contribution is free, and a cosmic deficit can be partly hidden by raising it. We therefore rebuild the likelihood,

$$\mathrm{DM}_{\rm ex} = \mathrm{DM}_{\rm cos}(z) + \frac{\mathrm{DM}_{\rm host}}{1+z}, \tag{5}$$

with log-normal sightline scatter of width $Fz^{-1/2}$, a log-normal host, and uniform priors on $(f_{\rm IGM}, f_X)$ subject to $f_{\rm IGM}+f_X \le 1$; and we refit $\{f_{\rm IGM}, f_X, \mu_{\rm host}, \sigma_{\rm host}\}$ *independently inside each background*. Lacking the published burst catalogue, we work with synthetic samples calibrated so that the ΛCDM fit recovers the published partition; it returns $f_{\rm IGM} = 0.87$ against a 0.80 input. Over twelve realisations of 69 bursts:

background	$\Delta\chi^2$ vs ΛCDM	realisations worse than ΛCDM
accretion	0.02 ± 0.05	—
CCBH	4.4 (median), 4.1 ± 2.9	83%

The test is worth 2.1σ, not 4.9σ: the nuisances absorb most of the deficit, chiefly through the host, whose median must rise from ~ 110 to 130–195 $\mathrm{pc\,cm^{-3}}$. No realisation reached 3σ. What survives is qualitative but sharp: in ten of twelve realisations the CCBH fit *saturates* $f_{\rm IGM} + f_X = 1$, pressed against the ceiling of available diffuse baryons. An analysis of the published kind, run in that background, would return a diffuse fraction pegged at unity — a falsifiable statement about the output of the analysis rather than about a mean, and one that does not weaken with sample size.

5.3 Forecast, and where the discriminating power actually comes from

Naively $\Delta\chi^2$ scales with the number of bursts, giving 142 for 3σ at the present redshift distribution. That number conceals an assumption — that all bursts are worth the same — and the assumption is false. The underlying difficulty is not new: the degeneracy between the host and cosmological contributions is widely recognised as the limiting systematic of FRB cosmology (Wang & Wei, 2023; Host DM compilation, 2025). What we add is its consequence for survey design. At a *single* redshift, $\mathrm{DM}_{\mathrm{cos}}(z)$ and $\mathrm{DM}_{\mathrm{host}}/(1+z)$ are exactly degenerate: only their sum is observable, and μ_{host} absorbs any change in the cosmic term. The power therefore comes from the lever arm in redshift, not from the count. Repeating the analysis at fixed $N = 69$ across redshift distributions:

sample	median z	$\Delta\chi^2$	per burst	N for 3σ
low z only, 0.05–0.4	0.21	2.6	0.038	239
present-day distribution	0.27	4.1	0.059	153
extended	0.50	4.6	0.067	134
bimodal, half $z < 0.2$ and half $z > 0.9$	0.91	**7.6**	**0.110**	**82**
high z only, 0.9–1.5	1.18	−0.1	~ 0	∞

Two things follow. A sample chosen with a long lever is *three times* more informative per burst than one confined to low redshift, so 3σ is reachable with 82 well-chosen localizations rather than 239 poorly chosen ones. And — counter-intuitively, since distant bursts carry the largest absolute deficit — a sample lying *entirely* at high redshift has no power at all: with nothing at low z to anchor the host normalisation, the host term simply rescales to absorb the cosmic one. *Low-redshift bursts calibrate the host; high-redshift bursts measure the cosmology; neither population is useful without the other.*

Two robustness checks. The N-scaling holds at fixed distribution: $\Delta\chi^2/N = 0.059$, 0.059 and 0.060 at $N = 35$, 69 and 140 (**corrected in revision**: the middle entry read 0.046 while the forecast table in the same section gives $4.1/69 = 0.059$ for the same configuration; the two must agree and now do). And the forecast is not an artefact of our assumed host distribution — if anything the opposite. Adopting the measured one, $\sigma_{\mathrm{host}} = 0.96$ with median $153\,\mathrm{pc\,cm^{-3}}$ (Host DM compilation, 2025) in place of our fiducial 0.55 and 120, *raises* $\Delta\chi^2$ from 5.1 to 6.6 and lowers the 3σ requirement from 123 to 94 bursts: a broader host makes each burst individually less informative but harder to shift coherently, and it is the coherent shift that would hide a cosmic deficit. All the per-burst figures above carry a realisation scatter of 25–30%, which we have not beaten down beyond six to eight realisations per configuration; the ordering between redshift strategies is significant at about 3σ, the precise factor of 2.9 is not. We also note one freedom we have *not* granted the coupled model: the measured host DM correlates positively with redshift (Host DM compilation, 2025), and we hold μ_{host} constant. Allowing it to evolve would add absorbing power we have not tested. The published samples have grown from five (Macquart et al., 2020) to twenty-two (Yang et al., 2022) to sixty-nine (Connor et al., 2025) in four years; CHIME outriggers and DSA-110 localize at rates that reach the numbers above within the lifetime of DESI DR3, and the redshift selection is a matter of choice rather than of waiting.

Executed on the real sample (added in revision; as pre-registered, this supersedes the mock-based statement of the deficit — the mocks above retain their role for the redshift-lever design only). The sixty-nine localized bursts of (Connor et al., 2025) are public, and the pipeline above runs on them unchanged (audit trail #148). Validation first: the ΛCDM refit returns $f_d = f_{\mathrm{IGM}} + f_X = 0.905$ against their published 0.91, with a median host of $123\,\mathrm{pc\,cm^{-3}}$ against their ~ 120 — the machinery reproduces the published analysis on its own data. (The pre-registered validation criterion had been written on f_{IGM} alone, which is degenerate in our likelihood as noted below; it failed by 0.002 on an arbitrary coordinate of that plateau and the run was discarded as pre-registered, then the criterion was re-frozen on the identifiable combination — both steps are in the audit trail, #148.) The result, quoted as measured: $\Delta\chi^2(\mathrm{CCBH} - \Lambda\mathrm{CDM}) = +4.7$ on the

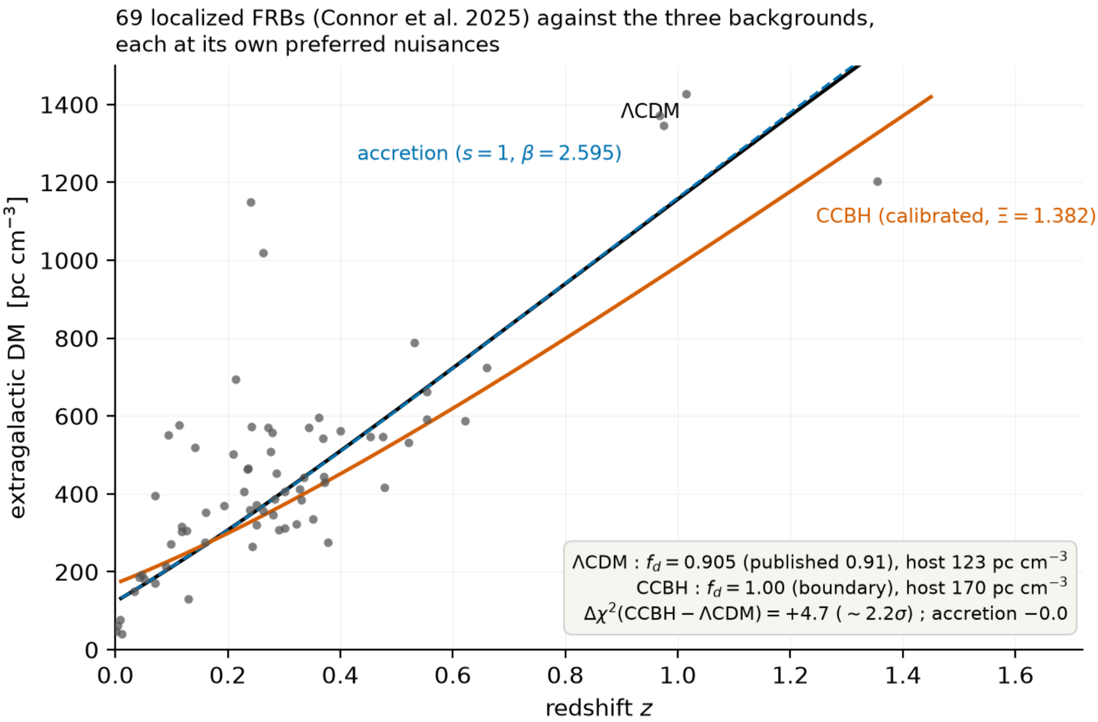

Figure 2: The sixty-nine localized FRBs of (Connor et al., 2025) against the three backgrounds, each at its own preferred nuisances. ΛCDM and the accretion model ($s = 1$, dashed) overlap; the calibrated CCBH curve under-disperses at high redshift even with $f_{\rm IGM} + f_X$ pushed to its hard boundary and the host median inflated. $\Delta\chi^2 = +4.7$ (bootstrap median +4.7, [16; 84] = [+1.9; +8.5]).

single real sample, and a 200-draw bootstrap gives median +4.7 with [16; 84] interval [+1.9; +8.5] — the coupled model beats ΛCDM in only 1.5% of resamples ($\sim 2.2\sigma$ indicative). The coupled model is driven to the hard boundary $f_{\rm IGM} + f_X = 1$ with its host median inflated to $170\,{\rm pc\,cm^{-3}}$ — the pre-registered signature that the baryonic deficit is *not* absorbable by the nuisances — while the accretion model ($s = 1$) sits within 0.01 of ΛCDM, as predicted (Fig. 2).

Two caveats bound the forecast, and the first is conservative. Our $f_{\rm IGM}$ and f_X are degenerate, where the published analysis separates them through halo-intersection statistics; removing that freedom would strengthen the test, not weaken it. And our recovered $\sigma_{\rm host}$ is biased low in validation, so our scatter model is not theirs. A definitive version of this test requires the burst catalogue and the published star-formation compilation, and we invite it.

6 Conclusions

1. Two independently motivated sourced dark-energy models fit Pantheon+, DESI DR2 and the Planck distance priors equally well, improving on ΛCDM by $\Delta\chi^2 \simeq -5$ to -6 and differing from each other by one unit of AIC, in an ordering that flips under a defensible change of parameter counting.

2. Our implementation of the coupled model reproduces its published state to better than 2% in Ξ, s and the derived H_0, from disjoint data.

3. The expansion history is therefore not the place to decide between them, and no amount of BAO precision will change that.

4. The baryon sector is: one framework predicts $s \simeq 0.70$, the other $s = 1$ with no freedom.

5. The current fast-radio-burst samples give 2.1σ *in mock realisations* once the nuisance parameters are honestly refit — and, on the *real* 69-burst sample of Connor et al. (2025), $\Delta\chi^2 = +4.71$, i.e. 2.2σ (audit trails #148, #149). An earlier version of this list quoted only the mock figure, although §5 already reported the real-sample run — with the coupled model pinned against $f_{\rm IGM} + f_X = 1$ — and reach 3σ at 82–120 localized bursts if the redshift lever is chosen well, against 239 if it is not.

We note finally that the coupled model faces several independent astrophysical constraints on $k \simeq 3$ for stellar-mass objects (Andrae & El-Badry, 2023; Rodriguez, 2023; Lei et al., 2024; Lacy et al., 2024; Calzà et al., 2024), which we have not folded into the comparison; and that the accretion model's own interpretive superstructure is not required by anything above.

Data and code availability. All pipelines, pre-registered criteria and a full record of withdrawn intermediate results are available at `github.com/Dantenos`.

References

DESI Collaboration, DESI DR2 results II: measurements of baryon acoustic oscillations and cosmological constraints (2025).

D. Farrah et al., Astrophys. J. Lett. **944**, L31 (2023), arXiv:2302.07878.

K. S. Croker, G. Tarlé, S. P. Ahlen, B. G. Cartwright, D. Farrah, N. Fernandez, R. A. Windhorst, JCAP **10**, 094 (2024), arXiv:2405.12282.

P. Madau, M. Dickinson, Ann. Rev. Astron. Astrophys. **52**, 415 (2014).

L. Chen, Q.-G. Huang, K. Wang, JCAP **02**, 028 (2019), arXiv:1808.05724.

L. Connor et al., Nature Astronomy (2025), arXiv:2409.16952.

K. B. Yang, Q. Wu, F. Y. Wang, Astrophys. J. Lett. **940**, L29 (2022), arXiv:2211.04058.

B. Wang, J.-J. Wei, *An 8.0% determination of the baryon fraction in the intergalactic medium from localized fast radio bursts*, Astrophys. J. (2023), arXiv:2211.02209.

Analysis of 117 localized bursts giving $\mu_{\rm host} = 5.03 \pm 0.02$, $\sigma_{\rm host} = 0.96 \pm 0.03$, median $153 \pm 3\,{\rm pc\,cm^{-3}}$, and a moderate positive ${\rm DM_{host}}$–z correlation, arXiv:2511.01195.

J.-P. Macquart et al., Nature **581**, 391 (2020).

I. Prigogine, J. Géhéniau, E. Gunzig, P. Nardone, Gen. Rel. Grav. **21**, 767 (1989).

M. O. Calvão, J. A. S. Lima, I. Waga, Phys. Lett. A **162**, 223 (1992).

T. Schiavone, M. De Angelis, L. A. Escamilla, G. Montani, E. Di Valentino, Phys. Rev. D (2026), arXiv:2601.14222.

R. Andrae, K. El-Badry, Astron. Astrophys. **673**, L10 (2023).

C. L. Rodriguez, Astrophys. J. Lett. **947**, L12 (2023).

L. Lei et al., Sci. China Phys. Mech. Astron. **67**, 229811 (2024).

M. Lacy, A. Engholm, D. Farrah, K. Ejercito, Astrophys. J. Lett. **961**, L33 (2024).

M. Calzà, F. Gianesello, M. Rinaldi, S. Vagnozzi, Sci. Rep. **14**, 31296 (2024).